MSCA Digitial Finance

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Lecture 2: Basics of Continuous-Time Portfolio Optimization

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Aims of this Lecture:

- Introduction to the main solution methods in continuous-time portfolio optimization
- Showing you how to cope with technicalities
- Allowing you to set up and solve continuous-time portfolio problems
- Going beyond standard problems

The Continuous-Time Portfolio Optimization Problem

For a given initial wealth of x>0 **decide** how many shares of which security to hold at what point in time to **maximize the investor's expected utility** from **final wealth at time T** (and from **consumption on [0,T]**).

⇨ Quite a complicated problem …

The Merton Problem

Consider a Black-Scholes-type market with a money market account and a stock via

$$dB(t) = B(t)rdt \qquad B(0) = 1, \quad \text{“MMA”}$$

$$dS(t) = S(t)\big((r+\lambda)dt + \sigma dW(t)\big) \qquad S(0) = s, \quad \text{“Stock”},$$

with W(t) a (one-dim.) Brownian motion.

This means that the price evolutions are given by

$$B(t) = e^{rt},\ S(t) = se^{\left(r+\lambda-\frac{1}{2}\sigma^2\right)t+\,\sigma W(t)}$$

with

$$E(S(t)) = se^{(r+\lambda)t},\ Var(S(t)) = s^2e^{2(r+\lambda)t}(e^{\sigma^2 t}-1)$$

$$P(S(t) \le x) = \Phi\left(\frac{\ln(x/s)-(r+\lambda-\frac{1}{2}\sigma^2)t}{\sigma\sqrt{t}}\right)$$

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What are the next steps?

- Describe the way how an investor acts at the above financial market
- Explain how to model the investor’s goal
- Find the best possible actions

Different descriptions of investment activities

We will describe the activities of our investor via different, equivalent concepts:

- **the trading strategy** $\varphi(t) = \big(\varphi_0(t), \dots \varphi_d(t)\big)'$ with

  $\varphi_i(t) =$ number of units of asset i held at time t

  where we require the pair of **trading and consumption** strategy $(\varphi(t), c(t))$ to be **self-financing**, i.e. the **wealth process** $X(t) = \varphi_0(t)B(t) + \sum_{i=1}^{d} \varphi_i(t)S_i(t)$ satisfies**:**

$$\textbf{(SF)} \qquad X(t) = x + \int_0^t \varphi_0(s)dB(s) + \sum_{i=1}^{d} \int_0^t \varphi_i(s)dS_i(s) - \int_0^t c(s)ds$$

- **the portfolio process** $\pi(t) = \big(\pi_1(t), \dots, \pi_d(t)\big)'$ with

  $\pi_i(t) =$ fraction of wealth invested in stock i at time t

  $1 - \sum_{i=1}^{d} \pi_i(t) =$ frac. of wealth in money market account at time t

Note that we have the following obvious relation between the two concepts:

$$\varphi_i(t)S_i(t) = \pi_i(t)X(t)$$

The Merton Problem – again

Consider a Black-Scholes-type market with a money market account and a stock via

$$dB(t) = B(t)rdt \qquad B(0) = 1, \quad \text{“MMA”}$$

$$dS(t) = S(t)\big((r+\lambda)dt + \sigma dW(t)\big) \qquad S(0) = s, \quad \text{“Stock”},$$

with W(t) a (one-dim.) Brownian motion.

Goal: Find a portfolio process $\pi * (.)$ (**fraction of wealth invested in the stock**) solving

$$v(0,x) := \sup_{\pi(.)\in A'(x)} E\left(U\left(X^{\pi}(T)\right)\right)$$

with wealth process

$$dX^{\pi}(t) = X^{\pi}(t)[(r+\pi(t)\lambda)dt + \pi(t)\sigma dW(t)],\ X^{\pi}(0) = x,$$

U(*x*) a utility function (i.e. twice differentiable, concave, strictly increasing),

$$\mathrm{A}'(\mathrm{x}) := \left\{\pi(t), t\epsilon[0,T] | \pi(.) \text{ prog. measurable}, X^{\pi}(t) \geq 0, E\left(U^{-}\left(X^{\pi}(T)\right)\right) < \infty\right\}$$

The Merton Problem – again

Note: For a constant portfolio process $\pi(t) \equiv \pi$ for all t we obtain

$$X^{\pi}(t) = x\, e^{\left(r+\pi\lambda-\frac{1}{2}\pi^2\sigma^2\right)t\, +\pi\sigma W(t)}$$

Why do we maximize expected utility? Where are the risk constraints?

Example:

Let $B \geq 0$ be a random variable, $C=E(B)$, U a utility function. Due to Jensen's inequality:

$$\Rightarrow E\big(U(B)\big) \leq U\big(E(B)\big) = U(C) = E\big(U(C)\big)$$

i.e. the utility function models a **risk-averse investor**, who prefers less risky payments given the same expectation.

How to obtain the appropriate utility function?

=> Your task (!), maybe some help later

Solution Method 1: Direct Verification – The Growth Optimal Portfolio

Solve $\sup_{\boldsymbol{\pi}(.)\in A(x)} E\left(\ln\left(X^{\pi}(T)\right)\right)$

Application of Itô's formula leads to

$$\ln(X^{\pi}(t)) = \ln(x) + \int_0^t \left(r + \pi(s)\lambda - \frac{1}{2}\pi(s)^2\sigma^2\right)ds + \int_0^t \pi(s)\sigma dW(s)$$

Under suitable assumptions on the portfolio process $\pi(s)$ we directly obtain the **growth optimal portfolio process** via maximizing the integrand in the dt-integral for every s∈[0,T] as

$$\pi^*(t) = \frac{\lambda}{\sigma^2}$$

or more general in a **d-stock setting** with even **random coefficients** as

$$\pi^*(t) = (\sigma(t)\sigma(t)')^{-1}\lambda(t)$$

Solution Method 1: Outlook – Further Applications

- Dealing with constraints on the portfolio process
  - Sustainability constraints (“How to obtain a sustainable portfolio?”)
  - Guarantee constraints (“Capital guarantee”)
- Using the GOP (“Growth optimal portfolio”) for
  - Constructing value preserving strategies
  - Serving as a benchmark
- Modelling the price evolutions for digital assets

Will be looked at in the final part of this lecture and in lectures of the next two days

Solution Method 2: The Martingale Optimality Principle and the HJB-Equation

Consider the following expected utility maximization problem

(1) $$\max_{u(.)\in A(x,I)} E^{0,x}\left(F\left(X^u(T)\right)\right),$$

where $A(x,I)$ denotes the set of **admissible control processes** on the time set I given that the initial state of the controlled process is x. The **utility function** $F(x)$ will be specified in the concrete applications.

To solve the problem (1), we look at a whole family of optimization problems given by the **value funtion**

(2) $$v(t,x) = \sup_{u(.)\in A\left(x,I(t)\right)} E^{t,x}\left(F\left(X^u(T)\right)\right)$$

where $I(t)$ denotes the restricted time set starting at time t. We will specify later how the process $X^u(t)$ is obtained and the set of admissible control processes is defined.

We will give a central result for solving stochastic control problems:

The Martingale Optimality Principle

If there exists a control strategy $u^*(.)$ such that with the definition of the function

(3) $$g(t,x) := E^{t,x}\left(F\left(X^{u*}(T)\right)\right)$$

we have that

(M) $g\left(t, X^{u*}(t)\right)$ is a martingale,

(SM) $g\left(t, X^{u}(t)\right)$ is a supermart. for all admissible control strategies $u(.)$,

then we obtain:

a) $u^*(.)$ is an optimal control strategy for problem (1).

b) For all possible initial states of the time and the controlled process (t, x) we get

(4) $$g(t,x) = v(t,x)\ ,$$

i.e. $g(., .)$ coincides with the value function of problem (1).

Consequence: Find such a control strategy $u^*(.)$

Application of the Martingale Optimality Principle

Let $X^u(t)$ be the unique solution of

(5) $\quad dX^u(t) = \mu\big(t, X^u(t), u(t)\big)dt + \sigma\big(t, X^u(t), u(t)\big)dW(t), t \in [0,T],$

with $W(t)$ a one-dimensional Brownian motion, $u(.)$ an arbitrary **control process**.

Let us further assume that the function

(6) $\quad h(t,x) := E^{t,x}\left(F\big(X^u(T)\big)\right)$

is a $C^{1,2}$-function. Application of Itô's formula together with representation (5) leads to

(7) $h\big(t, X^u(t)\big) = h(0,x) +$

$$+ \int_0^t \big[h_t\big(s, X^u(s)\big) + h_x\big(s, X^u(s)\big)\mu\big(s, X^u(s), u(s)\big) + {}^1\!/_2\, h_{xx}\big(s, X^u(s)\big)\sigma^2\big(s, X^u(s), u(s)\big)\big]ds$$

$$+ \int_0^t h_x\big(s, X^u(s)\big)\sigma\big(s, X^u(s), u(s)\big)dW(s).$$

Application of the martingale optimality principle Theorem 1 now yields the following

Theorem 2: ***Verification theorem for the HJB-Equation***

Let there exist a polynomially bounded classical $C^{1,2}$-solution to the **Hamilton-Jacobi-Bellman Equation** of stochastic control

(8) $$0 = \sup_{u \in U} \{ g_t(t,x) + \mu(t,x,u) g_x(t,x) + {}^1\!/_2 \, \sigma^2(t,x,u) g_{xx}(t,x) \}$$

(9) $$g(T,x) = F(x) \ .$$

Then with $v(t,x)$ denoting the value function of problem (3) we have:

a) $$g(t,x) \geq v(t,x) \ \ \forall \ (t,x) \in [0,T) \times \mathbf{R} \ .$$

b) If there exists an admissible control $u^*(t)$ satisfying

$$u^*(t) \in \arg\sup_{u \in U} \{ g_t(t,x) + \mu(t,x,u) g_x(t,x) + {}^1\!/_2 \, \sigma^2(t,x,u) g_{xx}(t,x) \}$$

then we have

(10) $$g(t,x) = v(t,x) \quad \forall \ (t,x) \in [0,T) \times \mathbf{R},$$

and $u^*(t)$ is an optimal control strategy for problem (3).

(8) $$0 = \sup_{u \in U} \{ g_t(t,x) + \mu(t,x,u) g_x(t,x) + {}^1\!/_2 \, \sigma^2(t,x,u) g_{xx}(t,x) \}$$

(9) $$g(T,x) = F(x) \ .$$

Consequence of the Verification Theorem:

A three-step algorithm to solve the control problem

Step 1: (Formally) solve the optimization problem in the HJB-Equation (8) and obtain the candidate for the optimal control:

$\Rightarrow$ $u^*(t, x)$ (still depending on the unknown (!) value function and its derivatives).

Step 2: Insert $u^*(t, x)$ into the HJB-Equation (8), drop the sup-operator, solve the obtained partial differential equation with terminal condition (9) explicitly (if possible).

Step 3: Check all the assumptions made during Steps 1 and 2 (very important, often forgotten !).

Solving the Merton problem using the HJB-Equation – 1

In the Merton problem, we have: $\mu(t,x,\pi) = x(r+\pi\lambda)$, $\sigma(t,x,\pi) = x\pi\sigma$ and

(11) $$\max_{\pi \in A'(x)} E^{0,x}\left(\frac{1}{\gamma}\big(X(T)\big)^{\gamma}\right), \quad \gamma < 1, \gamma \neq 0 \text{ fixed},$$

with the value function

(12) $$v(t,x) = \max_{\pi \in A'(t,x)} E^{t,x}\left(\frac{1}{\gamma}\big(X(T)\big)^{\gamma}\right)$$

Corresponding HJB-Equation

(8) $$0 = \max_{\pi \in [a,b]} \left\{\frac{1}{2}(\pi\sigma x)^2 v_{xx}(t,x) + \big((r+\pi\lambda)x\big)v_x(t,x) + v_t(t,x)\right\}$$

(9) $$v(T,x) = \frac{1}{\gamma}x^{\gamma}$$

Step 1: "Solve the maximisation problem in the HJB-Equation"

(13) $$\pi^*(t,x) = -\frac{\lambda}{\sigma^2}\frac{v_x(t,x)}{x v_{xx}(t,x)} \quad .$$

Important: Note that we have implicitly assumed: $v_{xx} < 0, \pi*(t,x) \in [a,b]^n, v \in C^{1,2}$. (*)

Solving the Merton problem using the HJB-Equation – 2

Step 2: "Solve the resulting partial differential equation"

Put $\pi*(t,x)$ into equation (8), drop the sup-operator and obtain:

(14) $$0 = -\frac{1}{2}\frac{\lambda^2}{\sigma^2}\frac{(v_x(t,x))^2}{v_{xx}(t,x)} + rxv_x(t,x) + v_t(t,x), \quad v(T,x) = \frac{1}{\gamma}x^\gamma.$$

Ansatz:

(15) $$v(t,x) = \frac{1}{\gamma}x^\gamma f(t) \quad \text{for some suitable function } f(t).$$

$\Rightarrow$

(16) $$f'(t) = -\left(r\gamma + \frac{1}{2}\frac{\gamma}{1-\gamma}\frac{\lambda^2}{\sigma^2}\right)f(t), \quad f(T) = 1$$

$\Rightarrow$

(17) $$f(t) = exp\left(-\left(r\gamma + \frac{1}{2}\frac{\gamma}{1-\gamma}\frac{\lambda^2}{\sigma^2}\right)(T-t)\right)$$

(18) $$\pi^*(t,x) = \frac{1}{1-\gamma}\frac{\lambda}{\sigma^2}$$

Solving the Merton problem using the HJB-Equation – 3

Step 3: "Check the assumptions"

$v(t,x) = \frac{1}{\gamma} x^{\gamma} f(t)$ acc. to (17) satisfies (*), $\pi^*(t,x)$ satisfies (*) for suitable constants *a,b*.

Hence, choose them big enough and arrive at:

The optimal portfolio process is given by

(19) $$\pi^*(t,x) = \frac{1}{1-\gamma} \frac{\lambda}{\sigma^2}$$ **"Constant portfolio weigthts"**

Note the form of the optimal portfolio process and its dependence on γ, the risk aversion parameter!

Note also that the optimal wealth process has the form of

(20) $$X(t) = x \cdot exp\left(\left(r + \frac{1}{1-\gamma} \frac{\lambda^2}{\sigma^2} \frac{1-2\gamma}{2-2\gamma}\right) t + \frac{1}{1-\gamma} \frac{\lambda}{\sigma} W(t)\right)$$

In particular, it is strictly positive !

Solution Method 2: Outlook – More problems, more HJB-Equations

Modifications of the Merton problem: Including consumption

(21) $$dX^{\pi}(t) = X^{\pi}(t)[(r + \pi(t)\lambda)dt + \pi(t)\sigma dW(t)] - c(t)dt\,, X^{\pi}(0) = x$$

with consumption rate c(t) ≥ 0.

New problem:

(22) $$\max_{(\pi,c)\in A'(x)} E^{0,x}\left(e^{-\delta T}\frac{1}{\gamma}\big(X(T)\big)^{\gamma} + \int_0^T e^{-\delta t}\frac{1}{\gamma}\big(c(t)\big)^{\gamma}\,dt\right)$$

with discount rate $\delta > 0$ and

Corresponding HJB-Equation for the value function:

(23) $$0 = \max_{\pi\in[a,b],c\geq 0}\left\{\frac{1}{2}(\pi\sigma x)^2 v_{xx}(t,x) + \big((r+\pi\lambda)x - c\big)v_x(t,x) + v_t(t,x) + \frac{1}{\gamma}c^{\gamma} - \delta v(t,x)\right\}$$

(24) $$v(T,x) = \frac{1}{\gamma}x^{\gamma}$$

and many more variations …

Solution Method 2: Outlook – More problems, more HJB-Equations

One further modificaton: Merton problem with d stocks

(21*) $$dX^{\pi}(t) = X^{\pi}(t)[(r + \pi(t)'\lambda)dt + \pi(t)'\sigma dW(t)], X^{\pi}(0) = x$$

Same problem:

(22*) $$\max_{(\pi,c)\in A'(x)} E^{0,x}\left(\frac{1}{\gamma}\left(X(T)\right)^{\gamma}\right)$$

As the wealth equation is still **one-dimensional**, we get an only slightly modied corresponding HJB-Equation for the value function:

(23*) $$0 = \max_{\pi\in[a,b]^d, c\geq 0}\left\{\frac{1}{2}\pi'\sigma'\sigma\pi x^2 v_{xx}(t,x) + \left((r + \pi'\lambda)x - c\right)v_x(t,x) + v_t(t,x)\right\}$$

(24*) $$v(T,x) = \frac{1}{\gamma}x^{\gamma}$$

and many more variations …

Solution Method 3: The Martingale Method – Option pricing revisited

A result from option pricing (see e.g. K., K. (2001))

$$(*) \quad H(t) = exp\left(-\left(r+\frac{1}{2}\left|\left|\theta\right|\right|^2\right)t - \theta' W(t)\right),\ \theta := \sigma^{-1}(b-r1),\ 1 := (1,\dots 1)'$$

For every final F_T-measurable payment $B \geq 0$ with $x := E(H(T)B) < \infty$ there exists a replicating portfolio process $\pi(t) \in A(x)$ with wealth process $X^{\pi}(t)$ and

$$\mathrm{X}^{\pi}(\mathrm{T}) = \mathrm{B} \quad \text{a.s.}$$

Hence, **the only arbitrage free price** p_B of the contingent claim B is given by

$$p_B = E(H(T)B)$$

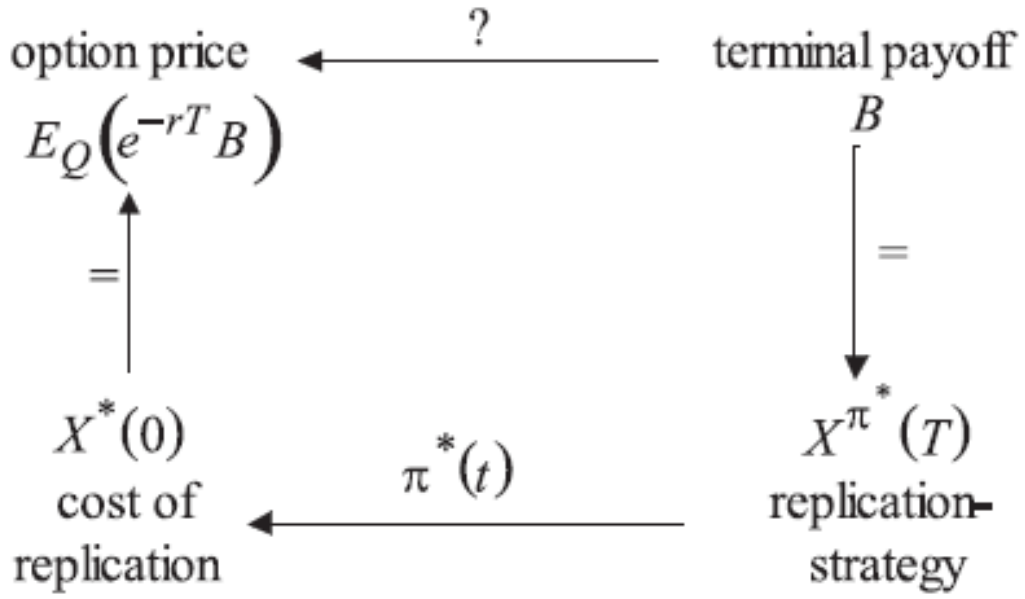
and the prices at time $t \in [0,T]$ are given by the **valuation process**

$$p_B(t) = E(H(T)B|F_t)$$

(this coincides with the usual formulation using an equivalent martingale measure).

Solution Method 3: The Martingale Method – Option pricing revisited

A graphical illustration of the result



Solution Method 3: The Martingale Method – Setting and Idea

We consider the following portfolio and consumption problem:

(25) $$\max_{(\pi,c)\in A'(x)} E^{0,x}\left(U_2\big(X(T)\big) + \int_0^T U_1\big(t,c(t)\big)\,dt\right)$$

where the two utility functions $U_2(.)$, $U_1(t,.)$ are assumed to be strictly concave, continuously on $(0,\infty)$ and satisfy the INADA-conditions

(26) $$U'(0) = \lim_{x\downarrow 0} U'(x) = +\infty, \quad U'(\infty) = \lim_{x\to\infty} U'(x) = 0\,.$$

Examples: $U(x) = \ln(x)\,,\ U(x) = \frac{1}{\gamma}x^{\gamma}, 1 > \gamma \neq 0$

Idea of the martingale method:

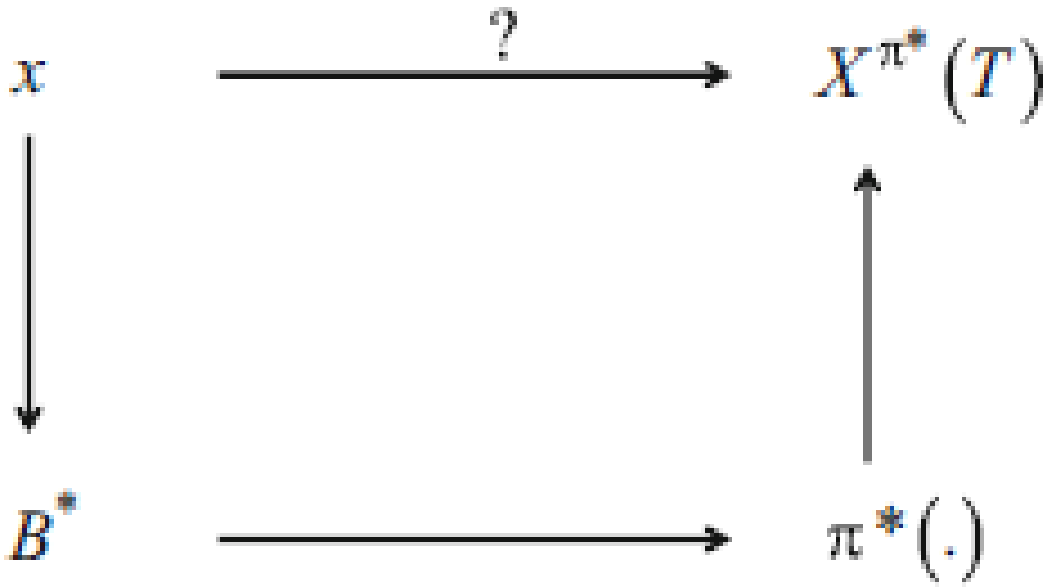
Decompose the dynamic optimization problem into

- A static optimization problem ("Determine the optimal consumption c(t) and the optimal final payment B=X(T)")
- A representation problem ("Determine the corresponding portfolio process π(t) such that $(\pi,c)\in A'(x)$ yield the wealth process X(t) with X(T) = B")

Solution Method 3: The Martingale Method – Graphical Idea

Given initial wealth x > 0

Get the corresponding optimal wealth process



Optimal payoff ("option") for the price of x

Determine the corresponding replication strategy for B*

Solution Method 3: The Martingale Method - Notations

We consider the following portfolio and consumption problem:

(25) $$\max_{(\pi,c)\in A'(x)} E^{0,x}\left(U_2\big(X(T)\big) + \int_0^T U_1\big(t,c(t)\big)\,dt\right)$$

where the two utility functions $U_2(.)$, $U_1(t,.)$ are assumed to be strictly concave, continuously on $(0,\infty)$ and satisfy the INADA-conditions

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Examples: $U(x) = \ln(x)\,,\ U(x) = \frac{1}{\gamma}x^{\gamma}, 1 > \gamma \neq 0$

Notations (needed for formulating the main theorem)

Let $I_2(y), I_1(t,y)$ be the inverses to the $U'_2(x), U'_1(t,x)$ and further

(27) $$H(t) = exp\left(-\left(r+\frac{1}{2}||\theta||^2\right)t - \theta' W(t)\right)$$

(28) $$\xi(y) = E\left(H(T)I_2\big(yH(T)\big) + \int_0^T H(t)I_1\big(yH(t)\big)dt\right)$$

(29) $$\Upsilon(x) = \xi^{-1}(x)$$

Solution Method 3: The Martingale Method – Optimality result

Theorem (Optimal consumption and optimal final wealth)

Let x > 0 and let $\xi(y) < \infty$ for all y>0. Then for

(30) $B^* = I_2\big(\Upsilon(x)H(T)\big)$ "optimal final wealth"

(31) $c^*(t) = I_1\big(\mathrm{t}, \Upsilon(x)H(t)\big)$ "optimal consumption rate"

there is a portfolio process $\pi^*(t)$ such that $(\pi^*(t), c^*(t))\epsilon A(x)$ such that we have

(32) $X^{\pi^*,c^*}(T) = B^*$

Note: We have (quite) **explicit formulas** for the **optimal consumption** and the **optimal final wealth**, but not for the optimal portfolio process.

Exercise:

a) Compute the optimal final wealth, portfolio and consumption process in the case of

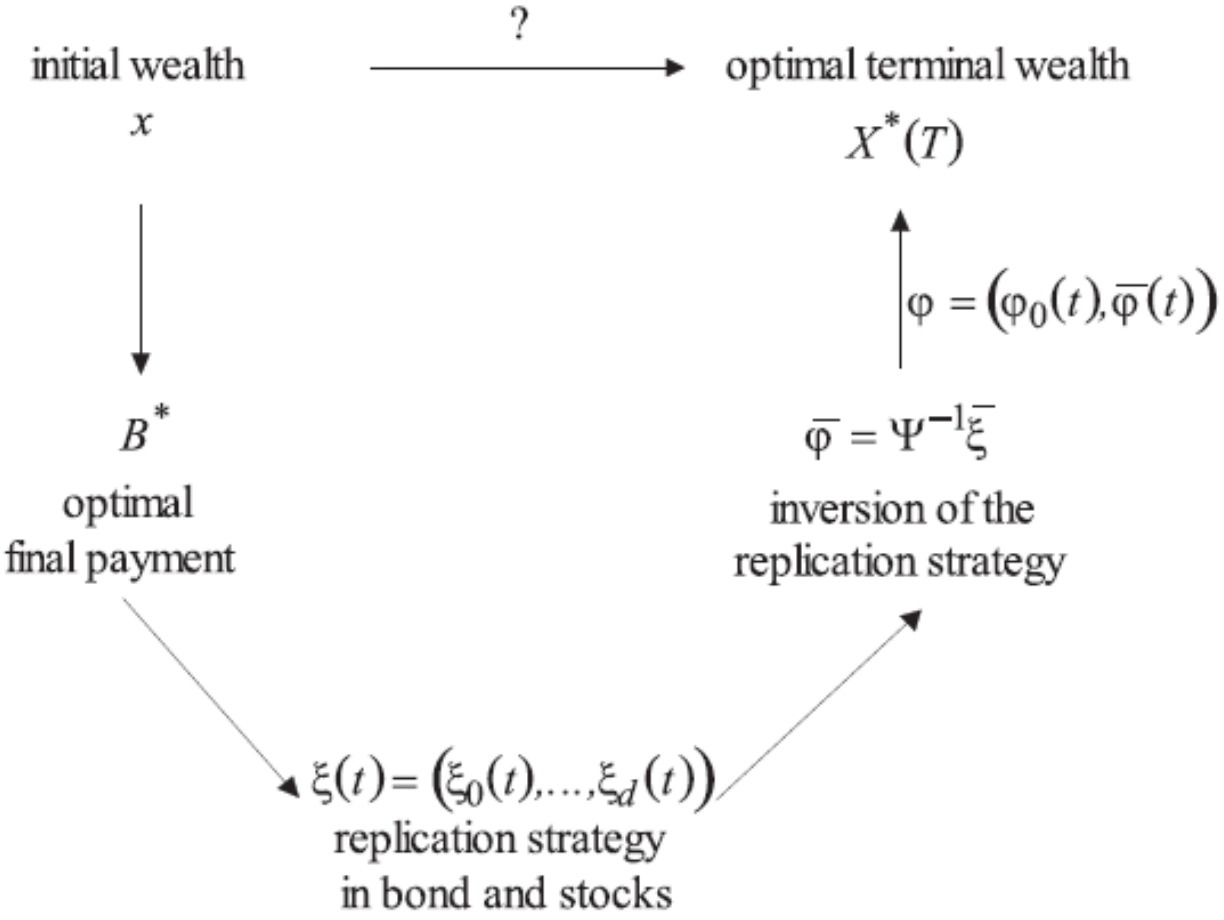
$U_2(x) = \ln(x)\,, U_1(t,c) = \ln(c)\,, t \in [0,1]$.

b) Under the assumptions of a), also determine the optimal portfolio process.

Solution Method 3: The Martingale Method – Applications

A popular application:

Optimal portfolios with derivative securities (see K., Trautmann (1998))



Solution Method 3: The Martingale Method – Optimal Option Portfolios

The Option Portfolio Problem:

Let us assume that we can trade in the **money market account** and ***d* options** on the *d* stocks with price processes

(33) $$f^{(i)}\big(t, S_1(t), \dots, S_1(t)\big) = \psi_{i0}(t)B(t) + \sum_{j=1}^{d} \psi_{ij}(t)S_i(t),\ i=1,\dots,d$$

where we have $f^{(i)} \in C^{1,2}$ and

(34) $$\psi_{ij}(t) = \frac{\partial}{\partial S_j} f^{(i)}\big(t, S_1(t), \dots, S_1(t)\big)\,,\ \psi_{i0} = \frac{f^{(i)}\big(t, S_1(t), \dots, S_1(t)\big) - \sum_{j=1}^{d} \psi_{ij}(t)S_i(t)}{B(t)}\,.$$

Note: This is in particular satisfied for European calls and puts in our model. Note also that for $f^{(i)}\big(t, S_1(t), \dots, S_1(t)\big) = S_i(t)$ we have our usual stock setting.

For a trading strategy $(\varphi_0(t), \dots, \varphi_d(t))$ in the money market account and *d* options, we consider the optimization problem

(35) $$\max_{\varphi \in A(x)} E\big(U(X^{\varphi}(T))\big)$$

with $U(.)$ a utility function and $A(x)$ the set of admissible trading strategies.

Solution Method 3: The Martingale Method – Optimal Option Portfolios 2

Theorem (Optimal Investment in Options)

Let the Delta matrix function $\Psi(t) = \left(\psi_{ij}(t)\right)_{i,j=1,\dots,d}$ (see (34)) be regular for 0≤t<T.

Then, the option portfolio problem (35) has the following explicit solution:

a) The **optimal terminal wealth** B^* **coincides** with the **optimal terminal wealth of the corresponding stock portfolio problem**.

b) Let $\xi(t) = \left(\xi_1(t), \dots, \xi_d(t)\right)$ be the optimal stock trading strategy in the corr. stock portfolio problem. Then, the optimal trading strategy in the option portfolio problem $\varphi(t) = \left(\varphi_0(t), \dots, \varphi_d(t)\right) =: \left(\varphi_0(t), \bar{\varphi}(t)\right)$ is given by

(36) $$\bar{\varphi}(t) = (\Psi(t)')^{-1}\xi(t)$$

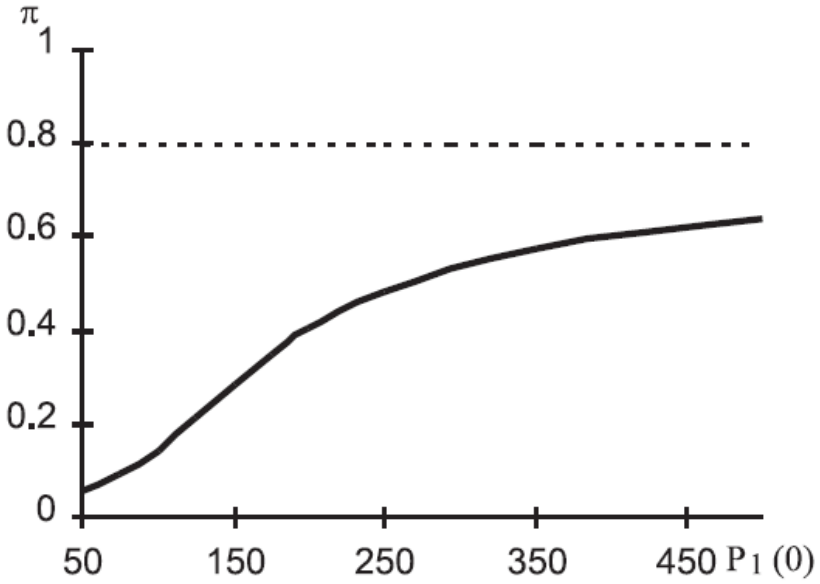
(37) $$\varphi_0(t) = \frac{1}{B(t)}\left(X(t) - \sum_{i=1}^{d} \varphi_i(t) f^{(i)}\left(t, S_1(t), \dots, S_1(t)\right)\right)$$

In particular, the optimal wealth process is identical to that in the optimal stock portfolio problem.

Solution Method 3: The Martingale Method – Optimal Option Portfolios 3

Interpretation: i) The (optimal final) **wealth** (process) in both the option and the stock portfolio problem depends only on the utility function, but is **independent** on the choice of the **tradable securities** under the assumptions of the theorem. ii) The optimal **trading strategy** **depends heavily** on the traded options via the **delta matrix** (i.e. via the replication strategy of the options).

Explicit example (see exercises)**:** U(x)=ln(x), d=1, option: European call option with strike K > 0, r = 0, b = 0.05, σ = 0.25



More examples and applications:

To come in the applications talks

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